

 CDI Semester 2 Final Exam Study Guide —
Simplified with Real Examples

1

 Programming & AI Development

Term	Definition	Real-life Example
Modifying Programs	Changing code to improve or fix it.	Fixing a Python app that crashes when you enter a wrong password.
Impact of Changes	The effects of changing a program (good or bad).	Changing a loop that accidentally causes a game to freeze.
Training	Teaching an AI model by feeding it lots of data.	Training a model with pictures of cats and dogs so it can recognize them.
Inference	Using the trained model to make predictions.	Uploading a new cat photo — the model says “This is a cat.”
Best Prompt (Role + Specificity + Structure + Constraints)	Writing a perfect question or command for AI to get accurate answers.	“As a history teacher, summarize World War I in 200 words.”
Prompt Engineering	Improving how you write prompts to get better AI results.	Changing “Explain GDPR” → “Explain GDPR in simple terms for beginners.”

2

 Software Development and Evaluation

Term	Definition	Real-life Example
Code Quality	How good and clean the code is.	Well-organized, commented code without bugs.
Software Evaluation	Judging how good a software is (speed, look, features).	Testing a new mobile app and giving it a rating.
Clarity	How clear and easy-to-understand the code is.	Using variable names like <code>totalScore</code> instead of <code>x</code> .
Iteration	Repeating steps to improve.	Updating a video game after player feedback.
Critical Thinking	Carefully analyzing problems.	Finding why an AI chatbot gives the wrong answers.

Term	Definition	Real-life Example
Linguistic Precision	Using exact words.	Instead of “stuff,” say “user data” when writing prompts.
AI Hallucinations	When AI makes up false info.	AI says “Elon Musk invented the iPhone.”
Document Summarization	Shortening long texts into main points.	Using ChatGPT to summarize a 10-page report.
Software Programming	Writing code to create programs.	Building a website in HTML/CSS/JavaScript.

3 Programming Concepts & Adaptive Systems

Term	Definition	Real-life Example
Programming Languages	Languages used to tell computers what to do.	Python, Java, C++, JavaScript.
Code Comparison	Comparing two codes to see which is better.	Comparing two sorting algorithms for speed.
Machine Learning	AI that learns from data.	Gmail spam filter learning from millions of emails.
Adaptive Systems	Systems that adjust to changes.	Netflix recommending new shows based on what you watch.

4 Ethics, Society, & Accessibility in Computing

Term	Definition	Real-life Example
Ethical Evaluation	Checking tech for fairness and moral impact.	Testing a face recognition app for racial bias.
Computational Artifacts	Anything created with computers.	Websites, apps, games, AI art.
Stakeholders	People affected by a system.	App users, developers, business owners.
Ethical Impact	How tech affects society ethically.	Facebook data leaks harming privacy.
Bias	Unfair outcomes in AI.	AI hiring tools favoring one gender.
Digital Divide	Unequal access to technology.	Some rural areas lacking internet.
Privacy	Controlling personal data.	Encrypted chats in WhatsApp.
Data Collection	Gathering info about users.	Websites collecting your browsing history.

Term	Definition	Real-life Example
Surveillance	Watching people using tech.	City cameras tracking car plates.
Misinformation	False information spreading.	Fake news on social media.
Accessibility	Making tech usable for everyone.	Screen readers for the visually impaired.

5 Equity & Global Access

Term	Definition	Real-life Example
Global Equity	Fair access to tech worldwide.	Giving free internet to underserved areas.
Computing Resources	The tools needed for computing.	Laptops, servers, internet bandwidth.
Global Society	A connected digital world.	Students from different countries collaborating online.
Technological Infrastructure	Tech systems that support digital life.	5G networks, fiber-optic internet.
Digital Literacy	Knowing how to use tech tools.	Teaching kids how to use spreadsheets.
Underserved Communities	Groups without good tech access.	Remote villages without internet access.

6 Computational Innovation & Emerging Tech

Term	Definition	Real-life Example
Computational Innovation	New creative uses of computing.	AI apps that turn sketches into full-color art.
Artificial Intelligence	Machines doing tasks that need human intelligence.	AI driving self-driving cars.
Big Data	Very large data sets used for insights.	Tracking customer habits in online shopping.
Telemedicine	Medical care via digital tools.	Online doctor appointments via video call.
EdTech	Educational technology.	Duolingo language learning app.
Virtual Reality	Computer-made 3D environments.	Using VR to train surgeons.
Renewable Energy Tech	Tech that supports green energy.	Smart solar panels.

Term	Definition	Real-life Example
Digital Transformation	Moving life & business online.	E-banking replacing physical bank visits.

7 Software Laws & Licensing

Term	Definition	Real-life Example
Software Laws	Rules about using and making software.	GDPR (privacy law in Europe).
Intellectual Property	Legal rights for creations.	Copyright for software code.
Copyright	Protecting original work.	Protecting a new video game you create.
Patent	Exclusive rights for inventions.	Patented AI algorithm for voice recognition.
Open Source Software	Free-to-use and modifiable code.	The Linux operating system.
GPL License	Open license — must share code.	GNU tools (like open-source compilers).
MIT License	Very flexible license — reuse allowed.	Many GitHub projects use MIT license.
Licensing Agreement	Legal terms of software use.	“I agree to the Terms of Service” checkbox.
Cybersecurity	Protecting systems from attacks.	Firewalls blocking hackers.
Encryption	Encoding data to protect it.	HTTPS for secure online shopping.
Breach	Unauthorized system access.	Hackers stealing credit card data.
Liability for Defects	Who is responsible if software fails.	Software bug causes bank errors — who pays?
Ethical Hacking	Legal hacking to find problems.	Pen testing for a bank's security.

8 Revision Questions (Examples)

Standard	Example Question
E-CDI-IC.25	How does cross-referencing reduce bias in AI results?
E-CDI-IC.25	What makes Copilot different from chatbots?
E-CDI-IC.25	What is AI vs what is not AI?
E-CDI-IC.25	Why tailor prompts for different departments?

Standard	Example Question
E-CDI-IC.26	What role does data play in AI?
E-CDI-IC.27	How have chatbots changed industries?
E-CDI-IC.27	How can AI improve video production?
E-CDI-IC.27	How can AI prompts boost sales?
E-CDI-IC.27	How does AI impact marketing & social media?
E-CDI-IC.27	Real examples of Narrow AI (Spotify recommendations).
E-CDI-IC.27	What is DALL-E and why is it useful for art?
E-CDI-IC.27	Why do artists use Midjourney?
E-CDI-IC.28	Can AI pass the Turing Test?
E-CDI-IC.28	Why is AI privacy important?
E-CDI.AP.22	Explain AI lifecycle: Training & Inference.
E-CDI.AP.23	What causes AI hallucinations?
E-CDI.AP.23	Why is prompt clarity important?
E-CDI.AP.23	Why is iteration important in prompt engineering?
E-CDI.AP.24	Relationship between AI and Machine Learning.

Answers:

E-CDI-IC.25

- How does effective cross-referencing contribute to identifying and mitigating bias in AI-generated outputs?** → It helps by comparing AI responses with reliable sources to detect incorrect or biased information, reducing the spread of bias.
- What distinguishes Copilot from other AI chatbots?** → Copilot integrates directly into software tools (like Microsoft Word or GitHub) to assist with tasks, while other chatbots focus on conversation.
- What is AI and what is not?** → AI is software that can perform tasks that usually need human intelligence, like language translation. A simple calculator is not AI because it follows fixed instructions.
- Why is it important to tailor AI prompts for different departments?** → Each department (HR, marketing, IT) needs different types of answers — tailoring prompts ensures the AI gives relevant, useful responses.

E-CDI-IC.26

- The role of data in powering AI** → Data is what AI learns from; good data helps AI make accurate predictions.

E-CDI-IC.27

How have AI chatbots influenced different industries? → AI chatbots handle customer service in retail, assist patients in healthcare, and provide legal info in law firms.

How can AI improve video creation processes? → AI can automate editing, add subtitles, or suggest improvements in videos.

How can AI prompts optimize sales performance? → AI can analyze customer data and suggest the best ways to approach leads.

How does AI transform marketing and social media management? → AI creates social media posts, analyzes trends, and improves targeting for ads.

How is generative AI transforming art and design? → It helps artists by generating ideas, patterns, and drafts quickly.

In what ways can AI support HR operations? → AI can screen job applications, schedule interviews, and answer employee questions.

In what ways is AI impacting music production? → AI can compose music, suggest improvements, or mix tracks.

Narrow AI vs. General AI → Narrow AI performs one specific task (like language translation). General AI would perform any human task (but it doesn't exist yet).

Real-life examples of Narrow AI → Siri, Google Translate, or spam filters in email.

What is the significance of DALL-E in creative AI applications? → DALL-E generates images from text descriptions, helping artists visualize ideas.

What makes Suno.ai valuable for musicians and creators? → Suno.ai helps musicians create new music or beats using AI.

What role does Runway ML play in AI-based video production? → Runway ML provides AI tools to edit videos, add effects, and create content.

Why are AI prompts valuable for product development teams? → They help teams quickly explore ideas and improve product design.

Why do we study AI? → To understand how it works, how it affects society, and how to use it ethically.

Why might artists choose Midjourney for their projects? → Midjourney generates artistic images based on text, helping artists explore new styles.

E-CDI-IC.28

Can AI become General: The Turing Test → The Turing Test checks if an AI can mimic human conversation so well that people can't tell the difference — passing this is a step toward General AI.

Why must data privacy and security be prioritized in AI applications? → AI uses personal data, so protecting it prevents misuse and builds trust.

E-CDI.AP.22

Best Prompt = Role + Specificity + Structure + Constraints → Example: "As a nutritionist, write a 300-word article about healthy diets for teenagers, using clear language."

Lifecycle of AI: Training and inference explained → Training: AI learns from data. Inference: AI uses what it learned to make predictions.

What is the 'Rule of Thumb' in prompt engineering? → Be clear and specific in your prompt to get better AI responses.

E-CDI.AP.23

How do examples of effective prompts enhance learning? → They show how to ask better questions, leading to clearer answers from AI.

How does AI improve document summarization? → AI can quickly shorten long documents into key points.

What causes AI hallucinations? → AI may make things up when it lacks correct data or the prompt is unclear.

What is the importance of mastering prompt engineering? → It helps you get more useful, accurate, and reliable AI outputs.

What role does AI play in software programming? → AI can suggest code, find bugs, or help write documentation.

What skills are developed through prompt engineering practice? → Clear thinking, communication, and understanding how AI works.

What skills are essential for creating AI-generated content? → Creativity, clear writing, and knowing how to guide the AI.

Why is clarity essential when crafting prompts? → Clear prompts prevent confusion and reduce AI errors.

Why is iteration important in prompt engineering? → Trying different versions of prompts improves the results.

E-CDI.AP.24

Relationship between AI and Machine Learning → Machine Learning is a part of AI — it's the method by which AI learns from data.